

MySQL Assignment 1 – DDL Commands & Constraints

Module: Data Analytics (DA) – Module 3

Project Overview

This report documents the creation and management of an Employee Database using MySQL DDL commands and constraints. The database includes Departments, Locations, and Employees tables with relationships and integrity constraints.

Objectives

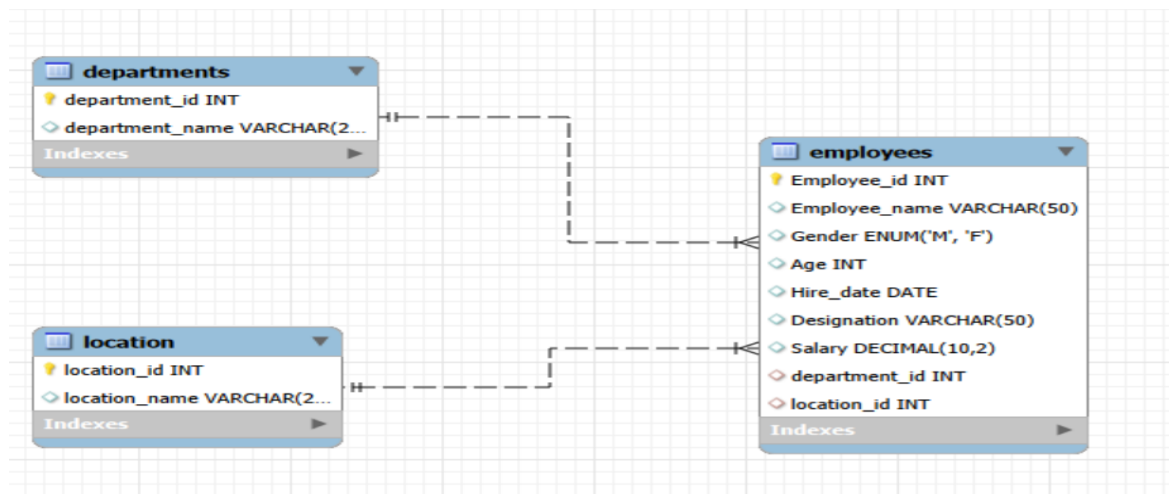
- Create a database and tables
- Apply DDL commands (CREATE, ALTER, RENAME, TRUNCATE, DROP)
- Implement PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, CHECK, DEFAULT, and AUTO_INCREMENT constraints.

Database Tables

Departments (department_id, department_name)

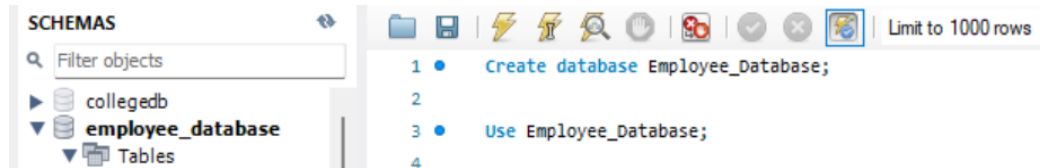
Locations (location_id, location_name)

Employees (employee_id, employee_name, gender, age, hire_date, designation, salary, department_id, location_id)

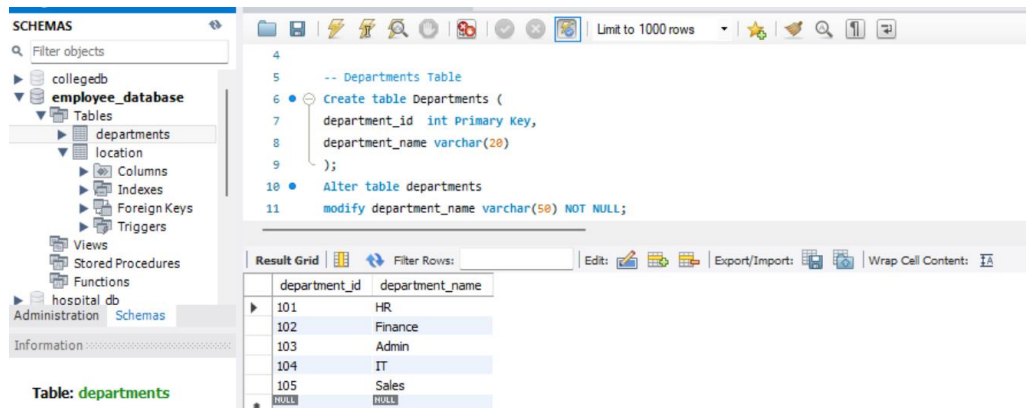


DDL Tasks Completed

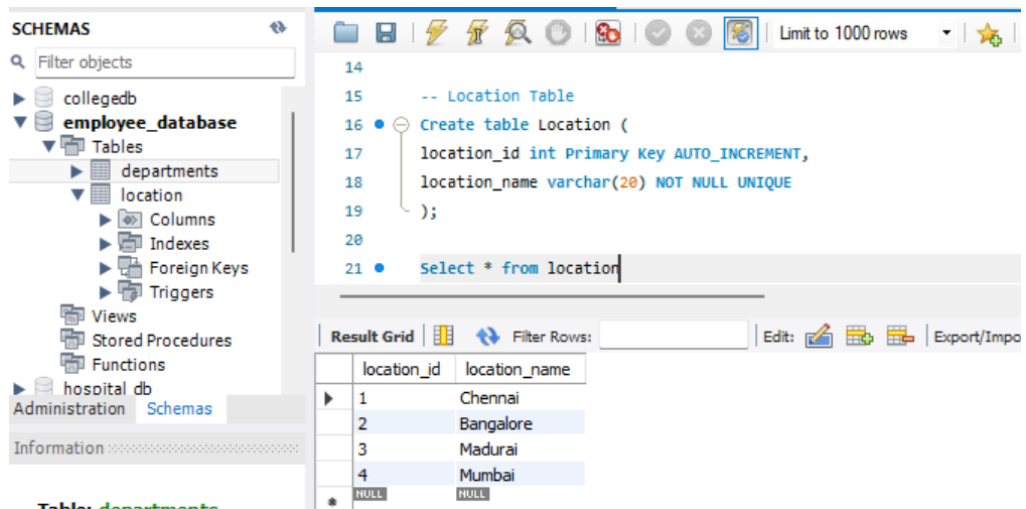
1. Created Employee_Database:



2. Created Departments Table:



3. Created Location Table:



4. Created Location Table:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Schemas' pane displays the 'employee_database' with tables 'departments' and 'employees'. The 'employees' table is selected, and its properties are shown in the 'Table: employees' pane. The 'Columns' pane lists the following columns and their data types:

- Employee_id: int PK
- Employee_name: varchar
- Gender: enum('M','F')
- Age: int
- Hire_date: date
- Designation: varchar
- Salary: decimal
- department_id: int
- location_id: int

The 'Result Grid' shows the following data:

Employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	department_id	location_id
201	Deepak	M	28	2024-01-15	Data Analyst	55000.00	103	1
202	Arun	M	25	2023-08-20	HR Executive	42000.00	101	2
203	Priya	F	28	2022-06-12	Accountant	50000.00	102	3
204	Karthik	M	31	2021-11-10	Sales Executive	48000.00	104	4
205	Divya	F	29	2020-03-18	Software Engineer	65000.00	103	2

5. Added email column:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Schemas' pane displays the 'employee_database' with tables 'departments', 'employees', and 'location'. The 'employees' table is selected, and its properties are shown in the 'Table: employees' pane. The 'Columns' pane lists the following columns and their data types:

- Employee_id: int PK
- Employee_name: varchar
- Gender: enum('M','F')
- Age: int
- Hire_date: date
- Designation: varchar
- Salary: decimal
- department_id: int
- location_id: int
- Email_id: varchar(100)

The 'Result Grid' shows the following data:

Employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	department_id	location_id	Email_id
201	Deepak	M	28	2024-01-15	Data Analyst	55000.00	103	1	NULL

6. Modified designation datatype:

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Schemas' pane displays the 'employee_database' with tables 'departments', 'employees', and 'location'. The 'employees' table is selected, and its properties are shown in the 'Table: employees' pane. The 'Columns' pane lists the following columns and their data types:

- Employee_id: int PK
- Employee_name: varchar(50)
- Gender: enum('M','F')
- Age: int
- Hire_date: date
- Designation: varchar(100)

The 'Result Grid' shows the following data:

Employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	department_id	location_id	Email_id
214	Ajay	M	32	2016-08-08	Senior				
215	Sneha	F	29	2020-09-08	Sales				

7. Dropped age column:

The screenshot shows a SQL IDE interface. On the left, the 'SCHEMAS' pane displays the 'employee_database' structure, including tables like 'employees', 'departments', and 'location'. The 'employees' table is selected, and its columns are listed: 'Employee_id' (int PK), 'Employee_name' (varchar), 'Gender' (enum), 'Hire_date' (date), 'Designation' (varchar), 'Salary' (decimal), 'department_id', and 'location_id'. The main editor shows the following SQL script:

```
86 • Alter table employees
87 Modify Designation varchar (100);
88
89 -- Drop Age column from the Employees table
90
91 • Alter table employees
92 drop column age ;
93
94
95
```

Below the script, the 'Result Grid' displays the data for the 'employees' table:

Employee_id	Employee_name	Gender	Hire_date	Designation	Salary	department_id	location_id
201	Deepak	M	2024-01-15	Data Analyst	55000.00	103	1
202	Arun	M	2023-08-20	HR Executive	42000.00	101	2
203	Priya	F	2022-06-12	Accountant	50000.00	102	3
204	Karthik	M	2021-11-10	Sales Executive	48000.00	104	4
205	Divya	F	2020-03-18	Software Engineer	65000.00	103	2
206	Rahul	M	2019-07-22	Manager	85000.00	101	1

8. Renamed hire_date to date_of_joining:

The screenshot shows a SQL IDE interface. On the left, the 'SCHEMAS' pane displays the 'employee_database' structure. The main editor shows the following SQL script:

```
93
94 -- Rename the column
95 • Alter table employees
96 change column Hire_date Date_of_joining date;
97
98
```

Below the script, the 'Result Grid' displays the data for the 'employees' table, with the 'Date_of_joining' column circled in red:

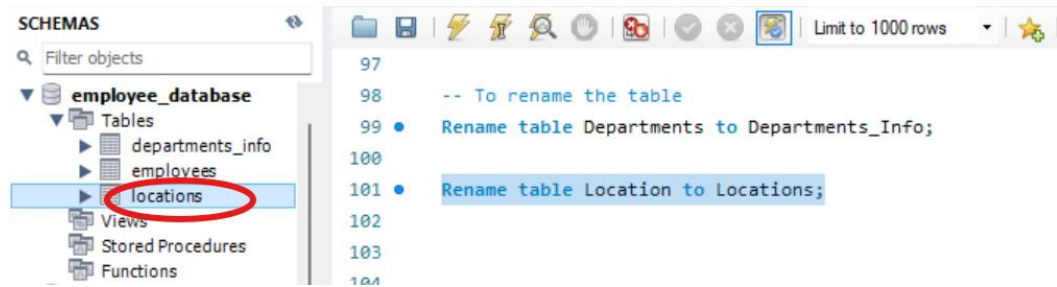
Employee_id	Employee_name	Gender	Date_of_joining	Designation	Salary	department_id	location_id
201	Deepak	M	2024-01-15	Data Analyst	55000.00	103	1
202	Arun	M	2023-08-20	HR Executive	42000.00	101	2
203	Priya	F	2022-06-12	Accountant	50000.00	102	3

9. Renamed Departments tables:

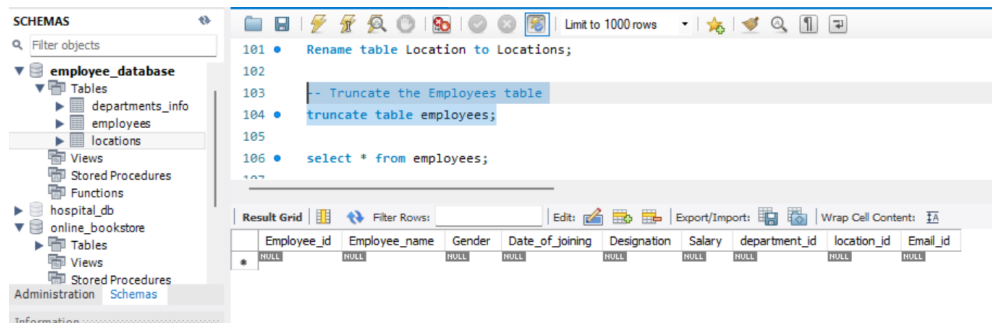
The screenshot shows a SQL IDE interface. On the left, the 'SCHEMAS' pane displays the 'employee_database' structure, with the 'departments_info' table circled in red. The main editor shows the following SQL script:

```
94 -- Rename the column
95 • Alter table employees
96 change column Hire_date Date_of_joining date;
97
98 -- To rename the table
99 • Rename table Departments to Departments_Info;
100
101
```

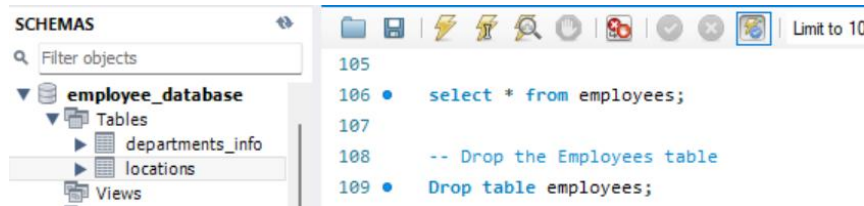
10. Renamed Location tables:



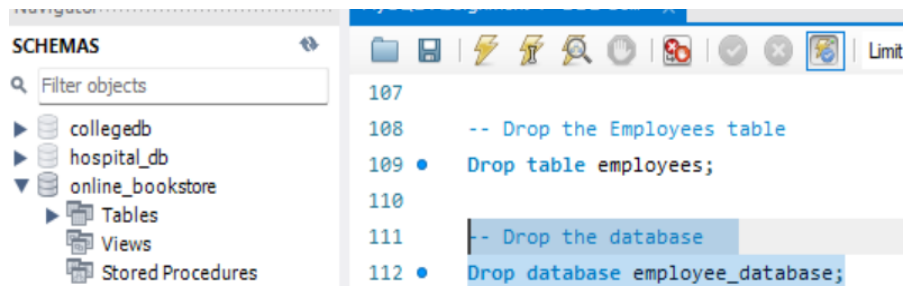
11. Truncated Employees table:



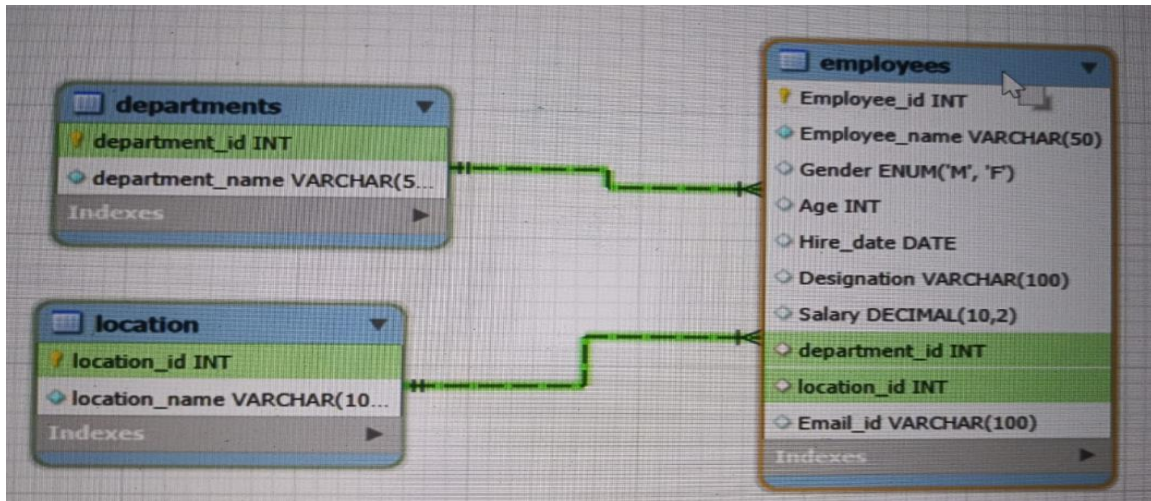
12. Dropped Employees table:



13. Dropped Employee_database:



14. Recreated schema with constraints:



Constraints

- PRIMARY KEY
- FOREIGN KEY
- NOT NULL
- UNIQUE
- CHECK (Age >= 18)
- DEFAULT CURRENT_DATE
- AUTO_INCREMENT

Conclusion

The assignment provided practical experience in designing relational databases, enforcing data integrity, and managing schema changes using MySQL DDL commands.